◎ Multivariate normal distribution

$$p(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right), \quad \Sigma = \Sigma^T, \quad |\Sigma| \neq 0$$

◎ Moment generating function (mgf)

$$M_x \equiv E[e^{t^T x}]$$

$$= \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \int \exp\left(t^T x - \frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right) dx$$

This integral can be written in the form of completing the square.:

$$(\text{The power index of the integrand}) = t^T x - \frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)$$

$$= \frac{1}{2}\left(-x^T\Sigma^{-1}x + x^T\Sigma^{-1}\mu + \mu^T\Sigma^{-1}x + 2t^T x - \mu^T\Sigma^{-1}\mu\right) \qquad (\because \Sigma = \Sigma^T)$$

$$= (\Sigma^{-1/2}x)^T(\Sigma^{-1/2}\mu + \Sigma^{1/2}t) + (\Sigma^{-1/2}\mu + \Sigma^{1/2}t)^T\Sigma^{-1/2}x$$

$$= -\frac{1}{2}\left[\Sigma^{-1/2}x - (\Sigma^{-1/2}\mu + \Sigma^{1/2}t)\right]^T\left[\Sigma^{-1/2}x - (\Sigma^{-1/2}\mu + \Sigma^{1/2}t)\right]$$

$$+ \mu^T t + \frac{1}{2}t^T\Sigma t$$

The gaussian integral formula is

$$\int \exp\left(-\frac{1}{2}(Ax+b)^T(Ax+b)\right) dx = \int \exp\left(-\left(\frac{Ax+b}{\sqrt{2}}\right)^T\left(\frac{Ax+b}{\sqrt{2}}\right)\right)\frac{2^{d/2}}{|A|} d\frac{Ax}{\sqrt{2}}$$

$$= \int \exp\left(-\left(\frac{Ax}{\sqrt{2}}\right)^T\left(\frac{Ax}{\sqrt{2}}\right)\right)\frac{2^{d/2}}{|A|} d\frac{Ax}{\sqrt{2}} = \frac{(2\pi)^{d/2}}{|A|}.$$

Therefore,

$$M_x = \exp\left(\mu^T t + \frac{1}{2}t^T\Sigma t\right) \quad \leftrightarrow \quad \mathcal{N}(\mu, \Sigma)$$

◎ Affine transformation

$$M_{Ax+b} = E[e^{t^T(Ax+b)}] = e^{t^T b}\, E[e^{(A^T t)^T x}]$$

$$= \exp\left(t^T b + \mu^T(A^T t) + \frac{1}{2}(A^T t)^T\Sigma(A^T t)\right)$$

$$= \exp\left((A\mu + b)^T t + \frac{1}{2}t^T A\Sigma A^T t\right) \quad \leftrightarrow \quad \mathcal{N}(A\mu + b, A\Sigma A^T)$$

Q Multiplication of normal distributions

$$\mathcal{N}(\mu_1, \Sigma_1)\, \mathcal{N}(\mu_2, \Sigma_2)$$

$$= \frac{1}{(2\pi)^d |\Sigma_1|^{1/2} |\Sigma_2|^{1/2}} \exp\left(-\frac{1}{2} (x-\mu_1)^T \Sigma_1^{-1} (x-\mu_1) - \frac{1}{2} (x-\mu_2)^T \Sigma_2^{-1} (x-\mu_2) \right)$$

The power of index is

$$-\frac{1}{2} (x-\mu_1)^T \Sigma_1^{-1} (x-\mu_1) - \frac{1}{2} (x-\mu_2)^T \Sigma_2^{-1} (x-\mu_2)$$

$$= -\frac{1}{2} x^T (\Sigma_1^{-1} + \Sigma_2^{-1}) x + \frac{1}{2} (\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2)^T x + \frac{1}{2} x^T (\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2)$$

$$- \frac{1}{2} \mu_1^T \Sigma_1^{-1} \mu_1 - \frac{1}{2} \mu_2^T \Sigma_2^{-1} \mu_2 . \quad \cdots (*)$$

The coefficient of the linear term can be written as

$$(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2) = (\Sigma_1^{-1} + \Sigma_2^{-1})(\Sigma_1^{-1} + \Sigma_2^{-1})^{-1} (\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2)$$

$$= (\Sigma_1^{-1} + \Sigma_2^{-1}) \left[(I + \Sigma_1 \Sigma_2^{-1})^{-1} \mu_1 + (\Sigma_2 \Sigma_1^{-1} + I)^{-1} \mu_2 \right]$$

$$= (\Sigma_1^{-1} + \Sigma_2^{-1}) \left[\Sigma_2 (\Sigma_1 + \Sigma_2)^{-1} \mu_1 + \Sigma_1 (\Sigma_1 + \Sigma_2)^{-1} \mu_2 \right] .$$

Then,

$$(*) = -\frac{1}{2} \left\{ x - \left[\Sigma_2 (\Sigma_1 + \Sigma_2)^{-1} \mu_1 + \Sigma_1 (\Sigma_1 + \Sigma_2)^{-1} \mu_2 \right] \right\}^T$$

$$\times (\Sigma_1^{-1} + \Sigma_2^{-1}) \left\{ x - \left[\Sigma_2 (\Sigma_1 + \Sigma_2)^{-1} \mu_1 + \Sigma_1 (\Sigma_1 + \Sigma_2)^{-1} \mu_2 \right] \right\}$$

$$+ \text{constants}$$